ChatGPT

The Topography of Synthetic Sexual Violence

Executive summary

Synthetic sexual violence is emerging as a distinct and rapidly scaling class of technology-facilitated sexual violence, enabled by generative AI systems that can fabricate or manipulate sexualized imagery, audio, and video with minimal friction, low cost, and increasingly high realism. [1]

Three recent, high-signal indicators illustrate the scale and trajectory of harm as of February 12, 2026. First, a January 22, 2026 analysis by the Center for Countering Digital Hate [2] found that an image-editing feature on X [3] led to an estimated ~3.0 million "sexualized" photorealistic images generated and posted over an 11-day period (Dec 29, 2025–Jan 8, 2026), including an estimated ~23,338 images that appeared to depict children; the study used sampling and an AI-assisted classifier calibrated against human labels. [4] Second, a February 4, 2026 statement by UNICEF [5] reported new survey-based evidence (Disrupting Harm Phase 2) conducted with ECPAT International [6] and INTERPOL [7] across 11 countries: at least 1.2 million children disclosed that their images had been manipulated into sexually explicit "deepfakes" in the prior year; in some countries, this equated to roughly 1 in 25 children. [8] Third, the Internet Watch Foundation [9] reported that, among AI-generated child sexual abuse videos it identified in 2025, 65% met its most severe "Category A" threshold; it also reported record volumes of confirmed CSAM reports overall. [10]

The "iceberg effect" is pronounced: the most measurable incidents (public posts, press-visible events, platform enforcement episodes) represent only a fraction of the underlying prevalence because of underreporting, stigma, fear of retaliation, uncertainty about legal remedies, and the technical reality that much generation and distribution happens in private channels or offline. [11] In the UK, one analysis integrating helpline and population estimates argued that for every woman supported for NCII abuse, ~53 others likely experience it without receiving assistance; the same source reported (via FOI responses from 37 police forces) a low conversion of complaints into charges/cautions. [12]

Displacement and escalation are visible in both criminal markets and behavioral adaptation. When platforms remove tools or content (e.g., bot/channel takedowns, feature restrictions), activity often moves to alternative platforms, encrypted spaces, delegated "deepfake-as-a-service" infrastructures, or local/offline generation where detection is significantly harder. [13] Simultaneously, technical sophistication is rising (diffusion-based video generation, face/voice cloning, automation, tokenized monetization), and cross-border harms are increasingly central—especially for sextortion and image-based abuse, where perpetrators and victims often reside in different jurisdictions. [14]

A key conclusion for researchers, policymakers, and platform designers is that no single control (forensic detection, watermarking, or criminalization alone) can address this harm class at population scale. Effective mitigation requires layered defenses: (1) product and platform safety-by-design ("abusability testing," friction, gating, and rapid response), (2) robust victim-centered reporting/takedown infrastructure, (3) cross-platform and cross-border coordination, and (4) measurement systems that explicitly model undercounting and the private-sphere shift. [15]

Definitions and scope

Synthetic sexual violence can be defined as the creation, manipulation, or use of synthetic (AI-generated or AI-altered) media to impose sexualized harm on a person or group without consent, including harms that functionally parallel sexual violence (humiliation, coercion, blackmail, reputational sabotage, and forced sexualized exposure), even when no physical contact occurs. This concept aligns with definitions of technology-facilitated sexual violence and technology-facilitated gender-based violence used in public-sector and UN framing. [16]

Technology-facilitated gender-based violence has been defined by UNFPA [17] as violence committed, assisted, aggravated, or amplified via information and communication technologies, encompassing sextortion, image-based abuse, cyberstalking, and grooming, among other forms. [18] A public-sector taxonomy similarly defines technology-facilitated sexual violence as misuse of technology to impose sexual conduct without consent (including deepfakes and sextortion). [19]

Key term boundaries used in this report:

Deepfakes: Media generated or manipulated with AI designed to look real (image/video/audio). UNICEF [5] explicitly describes deepfakes as AI-generated or AI-manipulated images/videos/audio designed to look real, noting their increasing use in sexualized content involving children (including “nudification”). [20] The term “deepfake” gained prominence from 2017 onward; a widely cited mapping report notes the term was coined by a Reddit user whose forum focused on face-swapping celebrity pornography. [21]

Non-consensual intimate imagery (NCII): The act or threat of creating, publishing, or sharing intimate imagery without consent (often colloquially called “revenge porn”), including content originally created consensually but later distributed without authorization. [22] In practice, NCII now includes AI-manipulated or AI-generated “nude” imagery where subjects never created intimate images at all. [23]

Revenge porn: A colloquial term describing non-consensual distribution of intimate images; many jurisdictions use different statutory terms (e.g., “intimate image abuse”). [24]

Sextortion: Extortion or coercion using sexual content or sexual threats. Public safety guidance distinguishes “financial sextortion,” which is increasingly prevalent among minors and typically involves threats to share sexual images unless payment is made; the reported offender geography is often cross-border. [25]

Synthetic child sexual content / AI-generated CSAM: Sexualized imagery of minors that is wholly AI-generated or AI-manipulated (including “nudification” or face-swaps of real children). The Federal Bureau of Investigation [26] has warned that CSAM created using generative AI or similar manipulation tools is illegal under U.S. federal law, including realistic computer-generated imagery. [27] The Internet Watch Foundation [9] reports demonstrate that offenders are already producing AI-generated child sexual abuse video at high volume and extreme severity. [10]

AI-facilitated abuse: Abuse in which AI systems materially reduce the cost, increase the scale, or expand the set of achievable harms (e.g., bulk nudification, automated targeting, model-driven persuasion, or synthetic evidence for coercion). [28]

Scope note: This report focuses on sexualized harms, victimization dynamics, measurement constraints, mitigation systems, and governance. It does not provide operational instructions for perpetration.

Iceberg effect, displacement, and escalation

The "iceberg effect" in synthetic sexual violence refers to the structural gap between (a) observable incidents recorded in platform data, law enforcement, helpline caseloads, and media coverage and (b) the true prevalence that remains unseen due to underreporting and technical invisibility. [29] This phenomenon is amplified by (1) the private nature of sexualized abuse, (2) victims' legitimate fear of reputational amplification ("reporting makes it spread"), and (3) the shift of distribution into private or encrypted channels where third-party measurement is harder. [30]

A concrete UK example illustrates the undercounting problem. One analysis combining population estimates and victim support volumes estimated that ~369,272 women experience NCII annually in the UK while the Revenge Porn Helpline assisted ~9,000 adult women in 2023—implying that for every woman supported, ~53 others may not receive assistance. [12] The same source reported FOI returns from 37 police forces (year 2024) showing 6,459 NCII-related complaints, 1,037 arrests, and 264 outcomes involving cautions or charges—highlighting attrition even among reported cases. [12]

Detection limits reinforce the iceberg. Hash-based systems (useful for known-illegal content) struggle with novel synthetic outputs because each generated variant has a different fingerprint. [31] Meanwhile, high-volume abuse can exist in direct messages, private groups, or bot-mediated workflows without meaningful public visibility. A WIRED [32] investigation documented large-scale usage of "nudify" bots on Telegram [33], describing how bot creators can reconstitute services after takedowns, suggesting a resilient ecosystem rather than isolated incidents. [34]

Displacement operates on multiple axes:

Platform displacement: CCDH's analysis describes a visible phase on X [3] followed by rapid restriction of access and technical changes after condemnation. [4] Research on cybercrime displacement in other digital markets suggests crackdowns may shift activity to other venues rather than eliminate demand, aligning with displacement theory observed in digital drug markets. [35]

Channel displacement into private/encrypted spaces: Sextortion investigations reported by entity["organization","National Center for Missing & Exploited Children","us child safety nonprofit"] and partners note movement of victims across platforms, including migration to end-to-end encrypted messaging apps. [36]

Geographic displacement: The Federal Bureau of Investigation [26] describes financially motivated sextortion as frequently cross-border, with offenders often located outside the U.S. (e.g., West Africa or Southeast Asia), complicating jurisdictional enforcement and victim support. [25]

Demographic and targeting displacement: Deepfake pornography has been empirically observed as overwhelmingly gendered (female targets) in early large-scale mapping; the same report also noted globalized targeting patterns (including non-Western victims in a substantial minority of cases). [21] Youth cases show diffusion into school contexts, where abuse often involves peers using commodified tools. [37]

Escalation is visible in three interacting trends:

Scale acceleration: CCDH's "Grok" event shows that when high-friction constraints are removed, output volumes can surge into the millions within days. [4] Large bot ecosystems similarly indicate industrialized throughput potential. [38]

Severity escalation: The Internet Watch Foundation [9] reports that AI-generated CSAM video is disproportionately represented in the most severe category tier compared to non-AI videos, suggesting AI is not merely replicating previous content distributions but enabling more extreme synthetic output. [10]

Market and automation escalation: Early mapping of deepfake ecosystems described commodification and monetization incentives (high view counts and specialized sites), and recent reporting on bots emphasizes token systems and potentially API-like provisioning that turns "deepfake creation" into a service layer. [39]

Threat models and enabling mechanisms

Timeline of key technical developments relevant to harms

Period	Capability shift	Relevance to synthetic sexual violence
2014	GANs introduced as a framework for training generative models. [40]	Establishes a foundational paradigm for realistic synthetic media generation, later adopted in face synthesis/manipulation pipelines. [40]
2017–2018	"Deepfake" term and early community diffusion via face-swapping pornography forums; later platform bans. [21]	Normalizes a "learned manipulation" affordance and creates a template for community-driven abuse tool ecosystems. [21]
2019	Large-scale mapping finds 96% of detected deepfake videos are non-consensual pornography; deepfake porn sites reach mass viewership. [21]	Establishes gendered threat baseline: sexualized abuse is primary use case, not fringe. [21]
2019	"Nudification" app visibility and replication dynamics (e.g., DeepNude) show rapid spread and copying even after shutdown. [41]	Demonstrates the "irreversibility problem": once a technique is released, clones persist. [42]
2021–2022	Diffusion models and latent diffusion approaches reduce compute cost and accelerate photorealistic generation; Stable Diffusion ecosystem expands tool access. [43]	Lowers barriers for high-quality text-to-image and image-to-image manipulation workflows used for nudification, face swaps, and synthetic pornography. [43]
2023	Training-data extraction from diffusion models demonstrated at scale (privacy leakage). [44]	Raises risk that models can regurgitate real people's images from training data, a pathway to re-victimization and non-consensual reproduction. [44]

Period	Capability shift	Relevance to synthetic sexual violence
2023–2025	Watermarking and provenance standards/tooling mature (C2PA, SynthID, robust watermarking research). [45]	Creates partial defensive infrastructure, but with adoption and stripping limitations. [46]
2024–2026	Large-scale incidents across mainstream and quasi-private ecosystems (Telegram bot markets; platform-integrated generation tools; survey-confirmed child victimization). [47]	Confirms movement from "edge cases" to systemic patterns impacting minors, schools, and public figures. [48]

Threat model diagrams

```
flowchart TD
  A[Target discovery] --> B[Input acquisition]
  B --> C[Generation / manipulation]
  C --> D[Distribution]
  D --> E[Victim impact]
  D --> F[Monetization / coercion]
  E --> G[Secondary harms]
  F --> G

  A:::stage -->|public figures, peers, partners| B
  B:::stage -->|social media photos, school sites, prior images| C
  C:::stage -->|text-to-image, image-to-image, face swap, voice clone| D
  D:::stage -->|public posts, DMs, groups, bot channels, marketplaces| E
  D:::stage -->|tokens, commissions, sextortion, paywalls| F
  E:::impact -->|reputation loss, trauma, isolation, safety risks| G
  F:::impact -->|repeat extortion, doxxing, offline stalking| G

  classDef stage fill:#eef,stroke:#66f,stroke-width:1px;
  classDef impact fill:#fee,stroke:#f66,stroke-width:1px;
```

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flowchart LR
  A[Model / tool provider] --> B[Platform integration]
  B --> C[User generation]
  C --> D[Content surfaces]
  D --> E[Detection & response]
  E --> F[Enforcement outcomes]
  F --> G[Displacement]
  G --> C

  E --> H[Victim reporting & support]
  H --> F
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classDef loop fill:#f7f7f7,stroke:#999,stroke-width:1px;
class A,B,C,D,E,F,G,H loop;
```

Technical mechanisms enabling AI-facilitated sexual abuse

Generative model capabilities: Modern diffusion models enable high-fidelity text-to-image and image-to-image transformations, including inpainting, style transfer, and “editing” workflows that can be weaponized for nudification or sexualized manipulation. [49] Earlier large-scale mapping emphasized that commodified tooling lowers barriers for non-experts, facilitating rapid scale. [21]

Face and voice manipulation: Synthetic sexual violence frequently relies on identity substitution (face swaps) or identity extension (voice cloning) to create “credible” evidence for harassment, humiliation, or coercion. The same general ecosystem of deepfake detection datasets exists precisely because face swaps are a recurring manipulation pattern. [50] For audio, benchmark datasets exist for spoofed speech and deepfake detection, reflecting the materiality of voice-based impersonation risks. [51]

Deepfake-as-a-service and commodification: Investigations into Telegram ecosystems describe bot-mediated workflows that require little technical skill, use token-like payment systems, and show resilience after enforcement (rapid re-creation). [52] Earlier market mapping found strong monetization signals in dedicated deepfake porn websites (very high view counts), indicating durable financial incentives rather than one-off trolling behavior. [21]

Model inversion and training data leakage: Privacy research shows two relevant classes of leakage. First, diffusion models can memorize and emit training examples, allowing extraction of real images from a model under some conditions. [44] Second, model inversion attacks for face recognition demonstrate that embeddings—often treated as privacy-preserving—can be used to reconstruct facial images; newer work even applies diffusion models in inversion pipelines. [53] In a sexual violence context, these findings raise the stakes of training data governance, consent, and downstream re-identification risks. [54]

Dataset poisoning: Poisoning has dual relevance. As an attack class, poisoning reflects the broader security surface of generative pipelines; as a defensive proposition, tools like Nightshade propose poisoning as leverage for content owners against scraping. [55] While not a direct “victim protection” mechanism for individuals at scale, poisoning research highlights adversarial dynamics that can shape both criminal and defense ecosystems. [56]

Evidence base and incident datasets

Case studies anchoring the current landscape

Platform-integrated generation shock: Center for Countering Digital Hate [2] documented rapid weaponization of a one-click editing feature on X [3], estimating millions of sexualized images generated and posted in days, including child-appearing images, and detailing the sampling and classifier calibration strategy used to estimate prevalence. [4]

Survey-confirmed child victimization: UNICEF [5] reported survey-based estimates across 11 countries involving ~1,000 internet-using children per country (ages 12–17) and ~1,000 caregivers, emphasizing that deepfake sexualization is not a niche harm but one already reported by large numbers of children. [20]

NGO forensic discovery and severity shift: Internet Watch Foundation [9] reports documented both rising quantities and high severity levels in AI-generated child sexual abuse video discovered in 2025. [10]

Criminal justice first-wave operations: Europol [57] operations (e.g., "Operation Cumberland") were described as among the first major cases targeting distribution of AI-generated child sexual abuse imagery, with investigators highlighting legislative gaps in many jurisdictions. [58]

Industrialized bot ecosystems: Investigations documented "nudify" bot ecosystems on Telegram [33] with millions of users, plus rapid reconstitution after takedowns—an archetype of displacement and resilience. [38]

School-based diffusion: Court and law enforcement actions show peer misuse in schools, including cases in Spain [59] and reports of charges in U.S. jurisdictions, illustrating low barriers and high harm even when perpetrators are minors. [37]

Measurement methodologies and the "how do we know" problem

Household surveys: UNICEF's Disrupting Harm Phase 2 description provides a model for representative measurement (multi-country, nationally representative sampling, standardized questions), but also underscores that surveys capture self-disclosed victimization and can still undercount due to shame or lack of awareness. [20]

Platform sampling + classifier estimation: CCDH's method—a random sample from a large population plus calibration against human labels and uncertainty intervals—illustrates one viable approach for quantifying platform-level output at high scale, though it may miss private messages and off-platform sharing. [4]

Helpline + FOI triangulation: UK-based analyses combining helpline caseloads, population baselines, and FOI disclosures provide one approach to model underreporting and justice-system attrition. [60]

CyberTipline data (and annotation ambiguity): NCMEC data streams provide high-volume indicators for online exploitation trends, but a public letter from Stanford's Center for Internet and Society details how "Generative AI" reporting fields can be semantically ambiguous and, in at least one prominent case, were widely misinterpreted by the public and policymakers—demonstrating that measurement pipelines can create narrative distortion if metadata is not carefully defined. [61]

Ethical constraints: Measurement for child sexual abuse material faces strict legal and ethical constraints; credible NGO investigations explicitly describe methods designed to avoid accessing illegal imagery and to minimize harm to analysts, aligning with safe research practices. [62]

Prioritized datasets and sources to consult

Priority	Source	What it enables	Access / notes
Highest	UNICEF Disrupting Harm Phase 2 (11-country surveys)	Prevalence estimates for child deepfake victimization; risk perception; cross-country comparability	National reports rolling out through 2026; statement provides methodology baseline. [20]
Highest	IWF annual and AI-focused reports	Verified counts and qualitative characterization of discovered (and actioned) CSAM, including AI-generated content severity	NGO analyst-confirmed datasets; not open raw media for obvious reasons. [10]
Highest	CCDH “Grok floods X…” study	A reproducible example of platform sampling methodology for synthetic sexual content outputs	Useful for measurement design; includes uncertainty intervals and calibration. [4]
Highest	NCMEC CyberTipline trend posts + cautionary analysis	Trend tracking across online enticement, sextortion, and AI-related reporting; highlights taxonomy pitfalls	Pair NCMEC numbers with methodological scrutiny (Stanford letter). [63]
High	Europol / law enforcement case summaries (e.g., Operation Cumberland)	Evidence of cross-border distribution networks and investigative barriers	Illustrates “first-wave” enforcement and statutory gaps. [64]
High	StopNCII and related partner documentation	Practical takedown and hash-sharing workflows for adult NCII; effectiveness claims	Privacy-preserving hashing approach; not for minors. [65]
High	Ofcom guidance and UK Parliament committee materials	A mature regulatory approach to “safety by design,” hash matching, and platform responsibilities	Useful for comparative governance and implementation design. [66]
Supporting	Deepfake ecosystem mapping reports (Deeptrace/Sensity)	Baseline proportions (sexual vs non-sexual), commodification signals, demographic patterns	Foundational baseline for “misuse is mostly sexual” claim. [21]
Supporting	Detection benchmark datasets (DFDC, FaceForensics++, ASVspoof)	Research evaluation for forensic detection models	Not specific to sexual content; useful for detector R&D. [67]

Legal and policy frameworks

Synthetic sexual violence sits at the intersection of criminal law (sexual offenses, harassment, extortion), civil remedies (privacy and image rights), platform governance (liability and moderation duties), and international cooperation (cross-border evidence and enforcement). [68]

United States: The TAKE IT DOWN Act (signed in 2025, per CRS summary) creates federal criminal prohibitions and establishes notice-and-removal requirements for covered platforms, with a one-year runway for platforms to implement required processes (deadline May 19, 2026 in the CRS overview). [69] Reporting also highlights free speech and due-process concerns raised by civil liberties groups, illustrating the governance trade space between rapid removal and risk of over-removal. [70]

European Union: Directive (EU) 2024/1385 on combating violence against women and domestic violence explicitly contemplates non-consensual production/manipulation (including by AI) of material making it appear that a person is engaged in sexual activities, where that material is made accessible to the public without consent; it also addresses threats and includes speech-right safeguards. [71] The regime foregrounds consent, victim protection, and harmonization goals while recognizing expression constraints. [72]

EU AI Act transparency: Article 50 (as summarized in accessible AI Act compendia) sets transparency obligations for synthetic content marking and deepfake disclosure, indicating an emerging regulatory expectation that deployers disclose manipulated content and providers ensure machine-readable marking of synthetic outputs; the cited resource ties obligations to a timeline extending into 2026. [73]

United Kingdom: Government press releases describe moves to criminalize the creation of sexually explicit deepfakes and to strengthen the legal handling of intimate image abuse, including deepfakes, alongside platform obligations under the Online Safety Act framework. [74] The UK regulator Ofcom [75] has issued “safety by design” guidance for women and girls online and has emphasized hash matching for intimate image abuse and other measures, while balancing privacy and expression considerations. [76]

Platform policies and enforcement reality: Enforcement gaps—especially in fast-moving, bot-mediated ecosystems—remain central. Telegram’s stated prohibition of nonconsensual pornography is documented alongside evidence of persistent bot ecosystems and rapid re-creation after takedowns. [52] Meta [77] has pursued legal action against vendors behind “nudify” apps advertised on its platforms, illustrating the shift from pure moderation to litigation and infrastructure disruption. [78]

Cross-border harms: Financial sextortion and distribution networks repeatedly implicate cross-border offender geographies, increasing the importance of mutual legal assistance, harmonized offenses, and standardized evidence handling. [79]

Detection, mitigation, and research agenda

Comparative table of detection and mitigation techniques

Technique	Typical accuracy profile	Scalability	False positive risk	Deployment complexity	Strengths	Failure modes / limits
Perceptual hashing of known content (e.g., PhotoDNA-style)	High for exact/near-exact known matches; not designed for novel variants [80]	Very high (cheap at scale) [81]	Low–medium (depends on thresholds and review) [61]	Medium (hash DB governance + review workflows) [82]	Excellent for repeat circulation of known illegal media [81]	Weak against novel synthetic outputs and small perturbations; needs strong governance and human review for edge cases [83]
Victim-centered hashing (StopNCII model)	Effective for blocking re-uploads of the same image/video via hashes [65]	High across participating platforms [84]	Medium (hash collisions unlikely; misreport risk exists)	Medium (cross-platform participation + user UX) [85]	Preserves privacy (image stays on device) and supports rapid mitigation [85]	Limited to adults and participating platforms; does not solve first-upload problem [86]
ML forensic detection (image/video/audio classifiers)	Variable; often fragile under distribution shift and adversarial adaptation [67]	Medium–high depending on model cost	Medium–high (context sensitive)	High (training, eval, adversarial testing) [50]	Can detect novel manipulations; supports triage at scale [50]	Arms race; false positives can harm victims/creators; requires careful evaluation, human review, and appeal processes [87]
Provenance / content credentials (C2PA)	Not "detection" of falsity; indicates origin and edits when preserved [88]	High in principle; depends on adoption	Low if signatures verify; usability errors possible	High (ecosystem rollout + UI surfacing) [89]	Cryptographic authenticity signaling; supports downstream trust systems [90]	Metadata can be stripped (e.g., screenshots, platform processing); adoption and consistent display remain limiting [91]

Technique	Typical accuracy profile	Scalability	False positive risk	Deployment complexity	Strengths	Failure modes / limits
Invisible watermarking (e.g., SynthID)	Effective for *own-model* outputs under expected transforms [92]	High on provider ecosystems; low cross-provider	Medium (tooling and detection access constraints)	Medium–high	Scales across a vendor's product suite; supports automated verification [93]	Vendor-specific; can be absent for open-source and many third-party models; research shows watermarks can be attacked/ removed in some settings [94]
Platform "safety by design" interventions (friction, gating, abusability testing)	Prevention-oriented; effectiveness depends on design and adversary adaptation [95]	High when built into product	Low direct FP risk; moderate indirect risk (overblocking)	Medium–high	Reduces mass abuse events; addresses "scale before detection" problem [96]	Can push abuse to other venues ("displacement"); must be paired with downstream mitigations [97]
Legal takedown and liability frameworks (e.g., TAKE IT DOWN)	Depends on enforcement and compliance mechanisms [23]	Medium (legal systems are slower than viral spread)	Medium (abuse of notice systems)	High (process + appeals + cross-border)	Creates strong incentives and standardized processes for removal [98]	Risk of over-removal, speech conflicts, jurisdictional gaps; requires transparency and safeguards [99]

Recommendations for research, policy, platform design, and victim support

Research and measurement priorities: Build measurement systems that explicitly model the iceberg effect, combining representative surveys (UNICEF-style), platform sampling studies (CCDH-style), and structured reporting/takedown datasets (StopNCII, helpline casework) while transparently documenting definitional boundaries and annotation ambiguity (as shown by the CyberTipline "Generative AI" reporting confusion). [100] Research programs should prioritize privacy-preserving methods, minimize exposure to illegal material, and publish ethics protocols for synthetic sexual violence measurement. [101]

Platform and product design priorities: Treat synthetic sexual violence as a predictable abuse case for any image/video editing or generative feature and require pre-deployment abusability testing, including "nudification" misuse scenarios and targeted harassment flows. [95] Invest in friction and gating (rate limits, identity/consent checks where appropriate, and rapid rollback capabilities) to prevent "millions-in-days" amplification. [102] Ensure robust user reporting and victim support escalation pathways, including cross-platform hash-based protections with strong privacy properties. [103]

Detection and provenance priorities: Deploy layered defenses rather than betting on a single technique. Hash matching remains essential for known CSAM and repeat content circulation, but must be complemented by (a) provenance systems (C2PA) and (b) watermarking where feasible, with explicit acknowledgment of stripping and removal attacks and the reality of open-source model ecosystems. [104] Fund independent red-teaming and adversarial evaluation for detectors and watermarks under real-world transformations and platform processing pipelines. [105]

Policy and legal priorities: Harmonize definitions and offenses for AI-mediated intimate image abuse and synthetic CSAM while maintaining clear safeguards for legitimate expression and research. [106] Expand victim-centered civil remedies and rapid takedown procedures with robust due-process protections, recognizing that “48-hour removal” systems can be both protective and vulnerable to abuse if not carefully designed. [23] Prioritize cross-border enforcement capacity where offenders are routinely offshore, and ensure victim services can coordinate across jurisdictions. [79]

Victim support priorities: Scale accessible, trauma-informed support mechanisms—including guidance for minors, school systems, and parents—because peer-produced and school-distributed deepfake nudes have become a recurrent pattern. [107] Support privacy-preserving takedown systems for adults (StopNCII) while ensuring specialized pathways for children, where legal and safeguarding requirements differ sharply. [108]

Gaps and future research priorities

Ground-truth scarcity and measurement bias: There is no comprehensive global dataset of synthetic sexual violence incidents because of ethical constraints, encryption, and offline generation. As a result, inference risk is high unless studies transparently model undercounting and sampling bias. [11]

Private-sphere governance: The most scalable harms are increasingly mediated by DMs, private groups, and bot ecosystems; a central research need is governance models that reduce harm without forcing unsafe trade-offs (e.g., undermining privacy for everyone). [109]

Model governance and training data accountability: Evidence that diffusion models can leak training images and that ambiguous “AI-related” reporting can distort policy discourse suggests a need for standardized, auditable training data governance, clear reporting taxonomies, and privacy risk evaluations tied to deployment contexts. [54]

Adversarial dynamics: Watermarking and provenance will remain contested; published work both proposes robust watermark methods and demonstrates removal attacks, implying an ongoing arms race. [110] Similarly, model inversion research indicates that “embeddings are safe” is not a stable assumption, affecting both biometric systems and downstream identity harms. [53]

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[1] [16] [19] https://opdv.ny.gov/technology-facilitated-gender-based-violence
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